

Chapter 11: Dynamic Vehicle Routing Problems

Vehicle Routing: Problems, Methods, and Applications
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Vehicle Routing Lecture Series

Slides prepared for self-study

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Introduction — Why Routing is Never Truly Static I

Real-world vehicle routing is rarely a “plan-once, execute-perfectly” problem. In most operational settings, the values of all input parameters are not known with certainty at the moment the planning horizon begins. Instead, information arrives *gradually* during the day: new customer requests are called in by phone or submitted via a mobile app, traffic jams develop unexpectedly, or a vehicle breaks down and must be taken out of service.

A classical, static Vehicle Routing Problem (VRP) assumes that all information is known *in advance* and *with certainty* before any vehicle leaves the depot. That assumption is violated in practice in at least three important ways:

- **Dynamic requests.** New customers may call during the day, requiring the dispatcher to insert them into already-running routes.
- **Dynamic travel times.** Congestion, road closures, and weather make travel times depend on the time of day and change unpredictably.
- **Dynamic vehicle availability.** Vehicles break down, become unavailable within certain windows, or return late from previous tours.

Introduction — Why Routing is Never Truly Static II

The field of **Dynamic Vehicle Routing** (DVR) addresses all three of these complications. The goal is to develop routing and dispatching algorithms that respond in real time to a changing environment whilst still producing high-quality, operationally feasible plans.

Introduction — Why Routing is Never Truly Static III

Why dynamic routing is harder than static routing

In a static VRP, the algorithm has unlimited time to compute an optimal plan because all data are fixed before execution begins. In a dynamic setting, the algorithm must (i) react to new information within seconds (because a waiting vehicle or a waiting customer is expensive), (ii) decide which actions to commit to *now* versus which to defer, and (iii) balance the quality of the current plan against the anticipated quality of plans that will be needed after future, as-yet-unknown events.

Key insight

The difficulty of dynamic VRP is not purely computational — it is also *informational*. The algorithm must reason about what it does not yet know and act under genuine uncertainty about the future.

Advances in technology have made dynamic routing increasingly tractable. GPS devices provide real-time vehicle positions; mobile and radio networks relay customer requests instantly; on-board computers accept re-routing instructions while vehicles are moving. These technologies have made dynamic dispatching systems commercially feasible and have stimulated a growing research literature.

Introduction — Why Routing is Never Truly Static IV

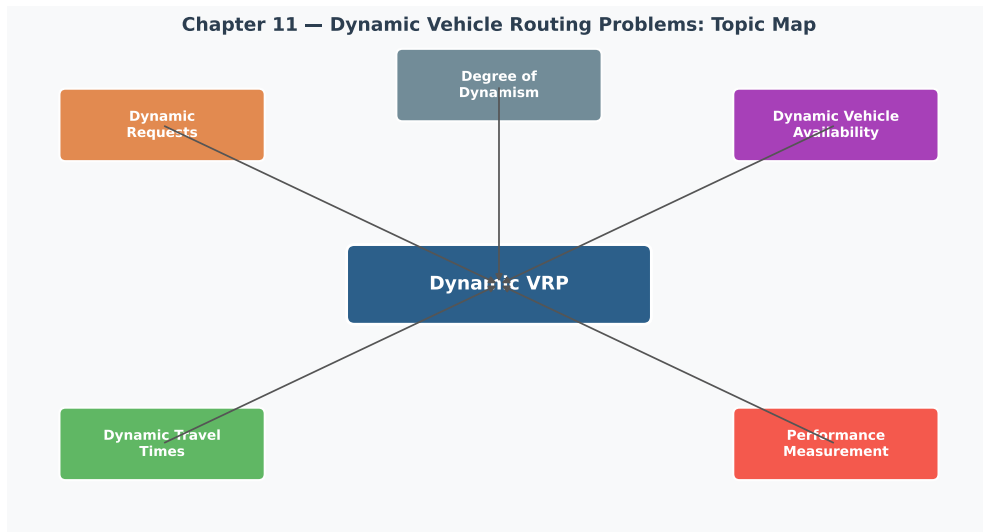


Figure: Topic map of Chapter 11. The chapter covers three main sources of dynamism (requests, travel times, vehicle availability) and discusses how to measure solution quality under uncertainty.

The Dynamic VRP — Formal Definition I

The chapter builds on the definition provided by Psaraftis (1995) and extends it to capture the modern operational landscape.

Definition: Dynamic VRP

A **Dynamic Vehicle Routing Problem** is a VRP in which some or all input parameters are not known at the beginning of the planning horizon but are *revealed progressively* while vehicles are executing their routes. The dispatcher must update the plan each time new information arrives, subject to operational constraints.

In mathematical terms, let time be indexed by $t \in [0, T]$ where T is the end of the planning horizon. At $t = 0$, a set of *a priori* (that is, known-in-advance) customers C^0 is available. During the horizon, additional customers C^t may arrive at random times $t > 0$. The full customer set at any time t is therefore $C^0 \cup C^t$. The dispatcher must assign all customers in $C^0 \cup C^T$ to vehicle routes whilst updating route assignments continuously as new customers arrive.

The Dynamic VRP — Formal Definition II

Key terminology

A priori customers: Requests that are known before the planning horizon begins; they can be included in an initial route plan.

Dynamic customers: Requests that arrive after the horizon begins; they must be inserted into existing routes in real time.

Committed actions: Actions that have already been *executed* by a vehicle and cannot be undone — for example, a delivery already made or a visit already begun.

Planned actions: Actions assigned to a vehicle in the current plan but not yet begun; these *can* be revised when new information arrives.

The distinction between committed and planned actions is central to dynamic routing: the dispatcher has freedom to re-plan future stops, but must not override stops that are already in progress.

The Dynamic VRP — Formal Definition III

Static vs. Dynamic Vehicle Routing

Static VRP — all customer locations known before departure



Dynamic VRP — new requests arrive while vehicles are en route



Figure: In a static VRP (top), all customer locations are fixed before any vehicle moves. In a dynamic VRP (bottom), new requests (red triangles) arrive while vehicles are already on the road, forcing real-time re-planning.

Sources of Dynamism — Three Categories I

The chapter identifies three distinct sources of dynamism that arise in real-world vehicle routing:

- 1. Dynamic requests** (Section 11.3 of the book). New customer service requests arrive during the planning horizon. A taxi dispatching system, a same-day parcel delivery service, or an emergency ambulance system all face this type of dynamism. The key challenge is to insert each new request into the current set of routes while disrupting existing commitments as little as possible.
- 2. Dynamic and time-dependent travel times** (Section 11.4 of the book). Travel times between locations change throughout the day due to congestion, traffic signals, road closures, and weather. This affects both route planning (which roads to take) and scheduling (when to arrive at each stop).
- 3. Dynamic vehicle availability** (Section 11.5 of the book). Vehicles may break down, may only become available after a certain time, or may need servicing mid-day. When a vehicle breaks down, its remaining customers must be redistributed to other vehicles, potentially requiring complete re-optimisation.

Degree of Dynamism I

Not all dynamic VRP instances are equally dynamic. To compare algorithms fairly across problem types, researchers have proposed various measures of how *dynamic* a problem instance is. The most widely cited measure is the **degree of dynamism** (dod), introduced by Larsen, Madsen, and Salomon (2002) and extended by subsequent authors.

Degree of Dynamism (dod) — Original Definition (Larsen 2002)

Let n_d denote the number of *dynamic* (online) customer requests that arrive after the planning horizon begins, and let n denote the *total* number of customers (both known and dynamic). The degree of dynamism is defined as:

$$\text{dod} = \frac{n_d}{n}$$

where n_d (read: “n sub d”) is the count of dynamically arriving requests and n is the total number of requests.

Degree of Dynamism II

Notation

Here n_d counts only the customers whose locations were *not* known when the initial plan was computed. n is the sum of all customers, both pre-known and dynamically arriving. The ratio dod (degree of dynamism) therefore falls in the interval $[0, 1]$: a value of 0 means a fully static problem (all customers known in advance), while a value of 1 means a fully dynamic problem (no customers known in advance at all).

How to read this formula

As dod increases, the problem becomes harder for *anticipatory* algorithms (which plan ahead) and easier for purely reactive algorithms (which only respond to what is already known). At very high dod values, there is little advance information to exploit, so sophisticated look-ahead strategies provide diminishing returns.

Degree of Dynamism III

Effective Degree of Dynamism (Larsen 2002, extended)

A criticism of the simple dod measure is that it ignores *when* dynamic requests arrive. A request arriving at the very start of the horizon is almost as useful as a known request, whereas a request arriving in the last few minutes is extremely disruptive. The **effective degree of dynamism** (edod) accounts for this by weighting each dynamic request by how late in the horizon it arrives:

$$\text{edod} = \frac{1}{n} \sum_{i=1}^{n_d} \frac{t_i^{\text{arrive}}}{T}$$

where t_i^{arrive} (read: “t sub i superscript arrive”) is the time at which dynamic request i becomes known, and T is the total length of the planning horizon.

Degree of Dynamism IV

Notation for edod

The sum runs over all n_d dynamic requests. The fraction t_i^{arrive}/T is a number between 0 and 1: it equals 0 if the request arrives at the very beginning and 1 if it arrives at the very end. Dividing the entire sum by n (total customers) normalises the metric so that it also lies in $[0, 1]$.

Intuition

The edod formula says: “on average, how late in the planning day do the dynamic requests arrive?” A high edod means most surprises come late, leaving less time to re-route, which makes the problem harder.

Degree of Dynamism V

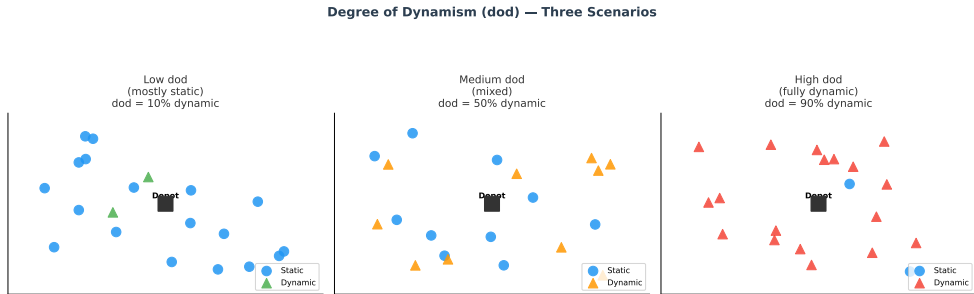


Figure: Illustration of the degree of dynamism across three problem instances. Blue circles are customers known before planning starts; coloured triangles are customers that arrive dynamically. Low dod (left) means most customers are known in advance. High dod (right) means most customers arrive in real time.

Degree of Dynamism VI

Objectives in Dynamic VRP

The primary objective in most dynamic VRP formulations is to minimise total travel distance (or total travel time) over all vehicles:

$$\text{Minimise } Z = \sum_{k=1}^K \sum_{(i,j) \in A} d_{ij} x_{ijk}$$

where K is the number of vehicles, A is the set of arcs (directed road segments between customer locations and the depot), d_{ij} (d sub i j) is the travel distance from node i to node j , and x_{ijk} (x sub i j k) is a binary decision variable equal to 1 if vehicle k travels directly from node i to node j on its route, and 0 otherwise.

Notation

K = number of vehicles in the fleet; A = set of all arcs (ordered pairs of locations); d_{ij} = distance (or time) from location i to location j ; $x_{ijk} \in \{0, 1\}$ = 1 if vehicle k uses arc (i, j) .

Other commonly used objectives include minimising the number of vehicles used, minimising the number of rejected (unserved) dynamic requests, and minimising waiting times for customers. A **hierarchical objective** may first minimise the number of vehicles and then minimise total distance.

Problem Variants — Taxonomy I

The chapter distinguishes between several broad categories of dynamic VRP:

Static and deterministic. All parameters are known and fixed. This is the classical VRP; it provides the benchmark against which dynamic approaches are compared.

Static and stochastic. All parameters are revealed before execution begins, but some of them (such as customer demand) are random variables. The algorithm plans once using probabilistic information.

Dynamic and deterministic. Parameters are revealed over time, but once revealed they are exact (no noise or uncertainty). For example, a new customer request arrives at time t with a precisely known location and demand.

Dynamic and stochastic. The most general case: parameters are revealed over time *and* involve uncertainty. For example, a customer may request service but their exact demand is only known upon vehicle arrival. Emergency medical dispatch systems often fall into this category.

Problem Variants — Taxonomy II

The focus of this chapter

The chapter concentrates on **dynamic and deterministic** problems because this is where the largest body of literature exists and where the distinction between reactive versus anticipatory approaches is clearest. Stochastic aspects are mentioned where relevant, particularly in discussions of anticipatory algorithms that use probabilistic models of future arrivals.

Common specific problem types discussed include:

- **Dynamic VRPTW**: VRP with Time Windows where both requests and time windows may be dynamic.
- **Dynamic Dial-a-Ride (DARP)**: pick-up and delivery of passengers, with dynamic pick-up requests and time-window constraints.
- **Dynamic PDP**: pick-up and delivery problems where goods must be picked up from one location and delivered to another.
- **Same-day delivery**: parcel delivery in which orders placed during the day must be delivered the same day.

Problem Variants — Taxonomy III

Information Flow in a GPS/Radio-Based Dispatching System

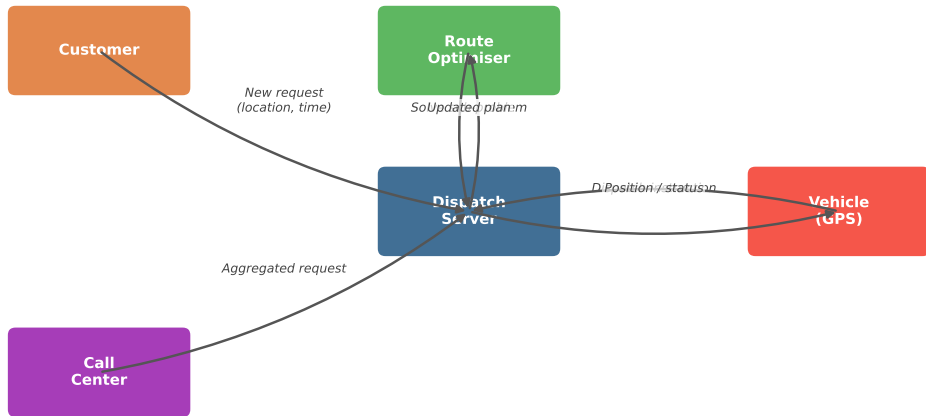


Figure: Information flow in a typical GPS-based dispatching system. Customers submit requests (upper left) or 20 / 85

Dynamic Requests — The Core Challenge I

The most studied form of dynamism in vehicle routing is the arrival of *new service requests* during the execution of a plan. Consider a courier company whose drivers have already left the depot with a pre-planned schedule. Throughout the day, new customers call in requesting delivery. Each new request must be assigned to a vehicle and inserted into that vehicle's remaining route — ideally without requiring the vehicle to travel far out of its way.

Dynamic Requests — The Core Challenge II

Problem statement: Dynamic VRPTW with online requests

At time $t = 0$, a set C^0 of a priori customers is known and an initial route plan is computed. During the planning horizon $[0, T]$, additional customers c arrive one by one at times $t_c > 0$. Each customer c has:

- a location (coordinates or distance from other nodes);
- a service time s_c (how long it takes to serve this customer);
- a time window $[e_c, l_c]$ where e_c (e sub c) is the earliest acceptable service time and l_c (l sub c) is the latest; and
- a demand q_c (quantity to deliver or pick up).

The dispatcher must insert c into a vehicle's route such that: (i) the vehicle's remaining capacity is not exceeded; (ii) all time windows (including those of already-scheduled customers) remain feasible; and (iii) the increase in total route cost is minimised.

11.3.1 Dynamic vs. Deterministic Problems — Solution Strategies I

The chapter classifies solution approaches for dynamic requests into three broad families: **reactive**, **periodic (rolling horizon)**, and **anticipatory** methods. Each family represents a different attitude towards the future.

Reactive methods treat each new request independently and insert it into the current best route immediately when it arrives. They make no attempt to reason about what future requests might look like. Reactive methods are very fast and simple to implement, making them popular in practice, but they can perform poorly when there are many future requests that could have been served more efficiently if the vehicle had been positioned differently.

Periodic (rolling horizon) methods re-optimize the complete route plan at regular time intervals — for example, every 5 minutes — rather than reacting to each individual request. This allows several new requests to be consolidated into a single re-planning step, which often produces better routes than inserting them one at a time.

Anticipatory methods explicitly model the uncertainty about future requests and use that model to make better decisions today. For example, rather than dispatching a vehicle immediately to a known

11.3.1 Dynamic vs. Deterministic Problems — Solution Strategies II

customer, an anticipatory method might hold the vehicle at a good waiting position for a short time in case a cluster of new requests arrives nearby.

11.3.1 Dynamic vs. Deterministic Problems — Solution Strategies III

Comparison of the three strategy families

Strategy	Reaction	Computation	Best suited for
Reactive	Per event	Very fast	High dod, time-critical
Periodic	Every Δt	Moderate	Mixed dod
Anticipatory	Per event + look-ahead	Expensive	Low-medium dod

Key insight

The choice of strategy should depend on the degree of dynamism. When most customers are known in advance (low dod), anticipatory methods that plan carefully for the future provide the largest improvement over a static baseline. When almost all customers arrive dynamically at short notice (high dod), the advantage of sophisticated anticipation shrinks and fast reactive methods become more competitive.

Insertion Heuristics for Dynamic Requests I

The simplest and most widely used reactive approach is the **cheapest insertion heuristic**: when a new customer c arrives, the dispatcher tries inserting c into every feasible position in every vehicle's current route, computes the extra cost of each insertion, and selects the insertion with the smallest additional cost.

Insertion Heuristics for Dynamic Requests II

Worked Example — Cheapest Insertion

Suppose vehicle 1's current route (not yet committed) visits customers in the order: Depot $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ Depot. A new customer N arrives. The distances (in km) are:

$$d(A, N) = 3, \quad d(N, B) = 2, \quad d(A, B) = 4, \quad d(B, N) = 5, \quad d(N, C) = 3, \quad d(B, C) = 2.$$

The extra cost $\Delta_{ij}(N)$ of inserting N between nodes i and j is:

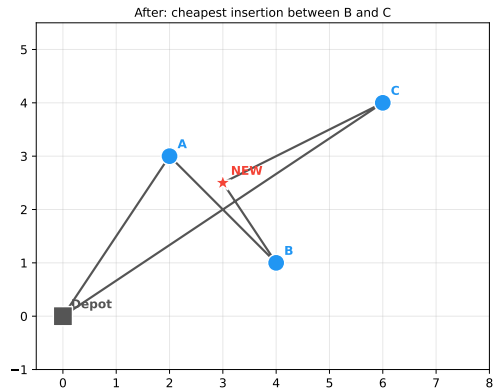
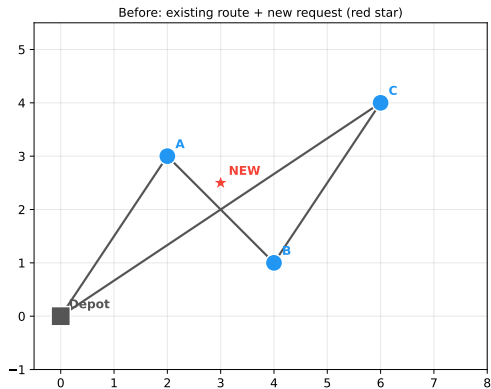
$$\Delta_{ij}(N) = d(i, N) + d(N, j) - d(i, j).$$

- Between A and B: $\Delta = 3 + 2 - 4 = 1$ km.
- Between B and C: $\Delta = 5 + 3 - 2 = 6$ km.

The cheapest insertion is between A and B (extra cost = 1 km), giving the new route: Depot $\rightarrow$ A $\rightarrow$ N $\rightarrow$ B $\rightarrow$ C $\rightarrow$ Depot.

Insertion Heuristics for Dynamic Requests III

Cheapest Insertion of a New Dynamic Request



Insertion Heuristics for Dynamic Requests IV

Figure: Cheapest insertion of a new dynamic request (red star) into an existing route. The algorithm evaluates inserting the new customer between every consecutive pair of nodes and selects the insertion with minimum added distance.

Insertion Heuristics for Dynamic Requests V

Pseudocode: Cheapest Insertion for a Dynamic Request

Input: Current route plan $R = (r_1, r_2, \dots, r_K)$ for K vehicles; new request c with location p_c , demand q_c , time window $[e_c, l_c]$.

Output: Updated route plan with c inserted.

- ① Set $\Delta^* \leftarrow +\infty$; best vehicle $k^* \leftarrow \text{none}$; best position $\text{pos}^* \leftarrow \text{none}$.
- ② **For each** vehicle $k \in \{1, \dots, K\}$:
 - ① Check remaining capacity: if $Q_k^{\text{rem}} < q_c$, skip this vehicle.
 - ② **For each** consecutive pair (i, j) in route r_k :
 - ① Compute $\Delta = d(i, c) + d(c, j) - d(i, j)$.
 - ② Check time-window feasibility of inserting c between i and j .
 - ③ If feasible and $\Delta < \Delta^*$: set $\Delta^* \leftarrow \Delta$, $k^* \leftarrow k$, $\text{pos}^* \leftarrow (i, j)$.
- ③ If $\Delta^* < +\infty$: insert c between i and j in route r_{k^*} .
- ④ Else: c cannot be served (reject or add emergency vehicle).

Insertion Heuristics for Dynamic Requests VI

Key insight

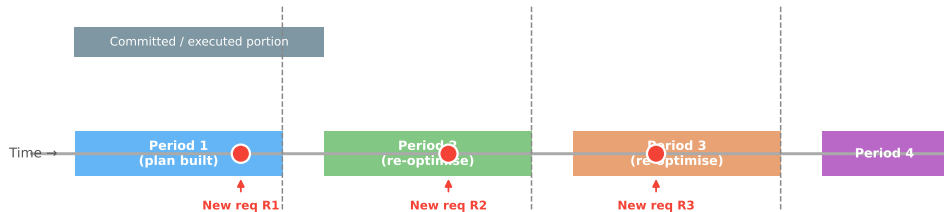
The time complexity of cheapest insertion is $O(K \cdot n)$ where n is the total number of stops across all routes. For typical fleet sizes this is fast enough to run in real time.

Periodic Re-optimisation — Rolling Horizon I

Instead of reacting to each individual request, a **rolling horizon** approach re-solves the complete routing problem at regular time intervals Δt . At each planning epoch (that is, each interval boundary), the dispatcher collects all requests that have arrived since the last re-optimisation and solves a new VRP that integrates both old (not-yet-served) and new customers.

Periodic Re-optimisation — Rolling Horizon II

Rolling Horizon / Periodic Re-optimisation in Dynamic VRP



Periodic Re-optimisation — Rolling Horizon III

Figure: Rolling horizon re-optimisation. The grey bar shows the portion of the route that has already been committed (vehicles are already en route to those customers). At each period boundary, new requests (red dots) are incorporated and the remaining plan is re-optimised.

Rolling Horizon — Algorithm

Input: Set of a priori requests C^0 ; re-planning interval Δt .

Output: Sequence of route plans $P^0, P^1, P^2, \dots$

- ① At $t = 0$: compute initial plan P^0 using all C^0 requests. Dispatch vehicles according to P^0 .
- ② **Repeat** at each epoch $t = \Delta t, 2\Delta t, 3\Delta t, \dots$:
 - ① Collect all new requests C^{new} that arrived in the interval $(t - \Delta t, t]$.
 - ② Freeze all vehicle actions that have already been committed (stops already begun or past their re-routing deadline).
 - ③ Re-solve the VRP on the remaining (uncommitted) customer set plus C^{new} , using current vehicle positions as new depots.
 - ④ Update the active plan; continue vehicle dispatch.

Trade-off: Δt choice

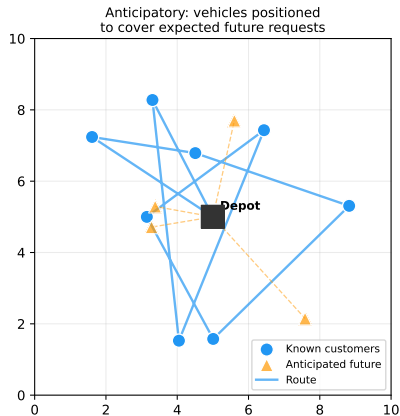
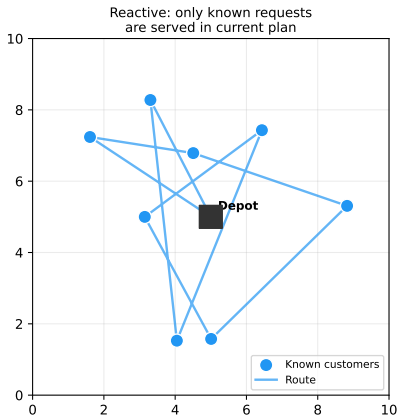
A small Δt (frequent re-planning) adapts quickly to new requests but uses more computational resources and may disrupt vehicle plans frequently. A large Δt is computationally cheap but may serve requests late or produce routes that are difficult to improve once many requests have accumulated. In practice, Δt between 5 and 30 minutes is common for courier applications.

Anticipatory Methods — Looking Ahead I

Anticipatory algorithms go beyond reacting to what is currently known. They use *probabilistic models* of future request arrivals to make better routing decisions today. The underlying logic is: “if I know that customers tend to appear in the eastern part of the city during the afternoon, I should position an idle vehicle there, even if no specific request has yet arrived.”

Anticipatory Methods — Looking Ahead II

Reactive vs. Anticipatory Routing Strategies



Anticipatory Methods — Looking Ahead III

Figure: Reactive routing (left) only plans for currently known customers. Anticipatory routing (right) also considers likely future request locations (orange triangles) and positions vehicles accordingly, reducing total travel distance when those requests materialise.

Anticipatory Methods — Looking Ahead IV

Key idea — Anticipatory routing

Anticipatory routing algorithms maintain a probabilistic model of future request arrivals. This model is typically a spatial-temporal distribution $\lambda(x, y, t)$ (lambda, read “lambda”) giving the probability density of a new request arriving at location (x, y) at time t . The algorithm then optimises not just over confirmed requests, but over the *expected future cost*, which accounts for both current requests and the likely cost of serving future requests.

Anticipatory Methods — Looking Ahead V

Expected Future Cost

Let $\mathbb{E}[C_{\text{future}}(\pi, t)]$ denote the expected cost of serving all future requests given that the current routing plan is π (pi, the Greek letter for the ratio of a circle's circumference to its diameter, here used as a general symbol for a plan) and the current time is t . An anticipatory algorithm selects the plan π^* that minimises:

$$\pi^* = \arg \min_{\pi} [C_{\text{current}}(\pi) + \gamma \cdot \mathbb{E}[C_{\text{future}}(\pi, t)]]$$

where $C_{\text{current}}(\pi)$ is the cost of serving currently known requests under plan π , and γ (gamma, a Greek letter used here as a weighting factor) controls the relative importance of future costs versus current costs.

Notation

π = current route plan (a complete assignment of customers to vehicles with a specified visit sequence);
 $C_{\text{current}}(\pi)$ = total distance driven to serve all currently known customers under plan π ; $\mathbb{E}[\cdot]$ = mathematical expectation (average over all possible future scenarios); γ = weight on future costs, typically chosen by cross-validation on historical data.

Routing Policies — No Wait, Wait, and Optimal I

A crucial decision in dynamic routing is what to do with a vehicle that has just finished serving its currently assigned customers and is waiting for new requests. Three broad policies are used:

No-Wait Policy

The vehicle immediately returns to the depot and waits there for the next dispatch instruction. This policy is simple and ensures capacity is always available, but it wastes travel time if new requests arrive in areas far from the depot.

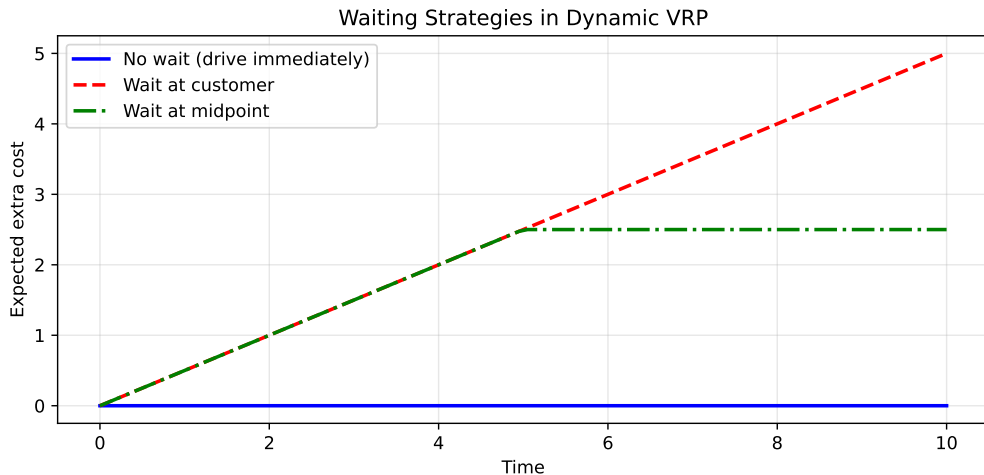
Wait-at-Last-Customer Policy

The vehicle remains at the location of its last customer until a new request arrives or until it is recalled. This avoids the return-trip cost but positions the vehicle at a location that may not be well-suited for future requests.

Optimal Waiting (Anticipatory Policy)

The vehicle moves to a *waiting position* that minimises the expected cost of serving future requests. The waiting position is computed by solving a sub-problem: “given the probability distribution of future requests, which location minimises expected additional travel?” Commonly, the optimal waiting position is the centroid (weighted average location) of the area where future requests are most likely to occur.

Routing Policies — No Wait, Wait, and Optimal III



Routing Policies — No Wait, Wait, and Optimal IV

Figure: Three waiting strategies for an idle vehicle (diamond marker). No-Wait (left): vehicle returns to depot. Wait-at-Last (centre): vehicle stays at the final served location. Optimal-Wait (right): vehicle moves to a position that minimises expected travel to future requests.

Routing Policies — No Wait, Wait, and Optimal V

Secomandi (2000) showed that the simple no-wait policy is often surprisingly competitive, especially when the depot is centrally located. However, for problems with clustered future request distributions, the optimal waiting position can substantially outperform both the no-wait and wait-at-last policies.

Key insight

The benefit of an intelligent waiting policy is largest when (i) travel times are high relative to service times, (ii) future requests cluster spatially away from the depot, and (iii) the algorithm has accurate information about where future requests are likely to appear. If the spatial distribution of requests is unknown or uniform, the no-wait policy performs as well as more sophisticated alternatives.

Sampling-Based Anticipatory Algorithms I

One of the most effective classes of anticipatory algorithms uses **Monte Carlo sampling** to generate multiple scenarios of future request arrivals and averages the optimisation results across those scenarios. This approach was pioneered by Bent and Van Hentenryck (2004) for the dynamic VRPTW.

Core idea — Multiple Scenario Optimisation (MSO)

At each decision point (that is, whenever a new request arrives or a vehicle becomes idle), the algorithm:

- 1 Generates S sample scenarios of future requests, drawn from a probabilistic model of arrivals.
- 2 For each scenario $s \in \{1, \dots, S\}$, solves the combined VRP on the union of known requests and the sampled future requests.
- 3 Selects the *immediate action* (which vehicle goes where next) that is best on average across all S scenarios.
- 4 Executes that immediate action; waits for the next event.

Sampling-Based Anticipatory Algorithms II

Mathematical statement

Let ω_s (omega sub s) be scenario s — a sampled set of future requests. Let $f(\pi, \omega_s)$ be the cost of plan π when future requests ω_s materialise. The algorithm selects the plan:

$$\pi^* = \arg \min_{\pi} \frac{1}{S} \sum_{s=1}^S f(\pi, \omega_s)$$

where the sum $\sum_{s=1}^S$ runs over all S sampled scenarios.

Notation

S = number of sample scenarios (typically 10 to 1000); ω_s (omega sub s) = one sampled realisation of future requests; $f(\pi, \omega_s)$ = objective value of plan π if scenario ω_s occurs; the fraction $1/S$ averages over all scenarios.

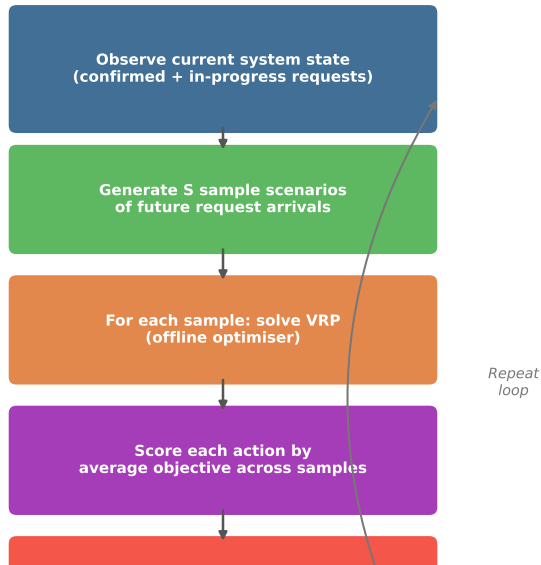
Sampling-Based Anticipatory Algorithms III

Why this works

By averaging over many scenarios, the algorithm identifies actions that are *robustly good* — they work well under a wide range of possible futures, not just under the current best guess. This hedging behaviour is what distinguishes anticipatory methods from purely reactive ones.

Sampling-Based Anticipatory Algorithms IV

Sampling-Based Anticipatory Algorithm for Dynamic VRP



Metaheuristics for Dynamic VRP I

Several metaheuristic algorithms originally developed for static VRP have been adapted to the dynamic setting. The adaptations typically involve running the metaheuristic in a time-limited fashion between events, and using warm-starts (initialising from the previous solution) to exploit the fact that plans change incrementally rather than completely.

Tabu Search for Dynamic VRP

Tabu search maintains a *tabu list* of recently visited solutions and forbids the search from returning to them. In a dynamic setting, tabu search is run continuously in a background thread. Whenever a new request arrives, it is added to the current best solution using cheapest insertion, and the tabu search then improves from that starting point.

The analogy: tabu search is like a hiker exploring a hilly landscape, deliberately avoiding recently visited locations (tabu list) to escape local optima and explore new terrain. In the dynamic setting, the landscape changes as new requests arrive, so the hiker must constantly adapt to new terrain whilst also trying to reach a better peak.

Evolutionary Algorithms for Dynamic VRP

Genetic algorithms (GA) maintain a **population** of candidate solutions. In the dynamic VRP context, an evolutionary algorithm:

- ① Maintains a population of feasible route plans.
- ② When a new request arrives, updates all population members by inserting the new customer at its cheapest position in each plan.
- ③ Applies crossover and mutation operators to generate new candidate plans.
- ④ Selects the best individuals for the next generation.

Because a diverse population is maintained, the algorithm naturally hedges against uncertainty: some plans may serve the new request cheaply, others position vehicles well for future requests.

Metaheuristics for Dynamic VRP III

Kilby, Prosser, and Shaw (1998) proposed a dynamic programming approach for the dynamic VRPTW. The state in their DP consists of the current vehicle positions and the set of pending requests. Transitions correspond to vehicle dispatching decisions. While the full DP is intractable for large instances, approximations and decompositions make it practical.

Dynamic Programming for Dynamic VRP — Bellman Equation

Let $V(s, t)$ denote the minimum expected total travel cost starting from state s at time t , where s encodes the positions of all vehicles and the set of unserved customers. The Bellman equation is:

$$V(s, t) = \min_{a \in A(s)} [c(s, a) + \mathbb{E}[V(s', t')]]$$

where $A(s)$ (A of s) is the set of feasible actions from state s , a is one such action (a dispatch decision), $c(s, a)$ is the immediate cost of action a , and s' is the next state reached after action a and any new arrivals during the transition.

Metaheuristics for Dynamic VRP IV

Notation

$V(s, t)$ = value function (minimum expected future cost from state s at time t); $A(s)$ = feasible action set; $c(s, a)$ = immediate travel cost of action a in state s ; $\mathbb{E}[V(s', t')]$ = expected future value after transitioning to state s' .

How to read the Bellman equation

The equation says: “the best policy from state s is to choose action a that minimises the sum of (i) what you pay right now and (ii) the expected best cost from the next state onward.” This recursive structure means that knowing V at the next step is sufficient to make a greedy optimal decision at the current step — this is the *principle of optimality*.

Python — Cheapest Insertion for Dynamic Request

The following Python function implements the cheapest-insertion rule for adding a new dynamic customer into an existing set of routes. The function checks capacity and time-window feasibility at each candidate insertion position and returns the best (minimum extra cost) insertion.

```
import math
from typing import List, Tuple, Optional

def distance(p1: Tuple[float, float], p2: Tuple[float, float]) -> float:
    """Euclidean distance between two (x, y) coordinate pairs."""
    return math.hypot(p1[0] - p2[0], p1[1] - p2[1])

def cheapest_insertion(
    routes: List[List[dict]],          # each route is a list of customer dicts
    new_customer: dict,                # {'id', 'x', 'y', 'demand', 'tw_open', 'tw_close'}
    vehicle_capacities: List[float],
    vehicle_loads: List[float],
) -> Optional[Tuple[int, int]]:
    """
    Find the cheapest feasible insertion of new_customer into routes.

    Returns (route_index, insert_after_index) or None if infeasible.
    insert_after_index = i means insert new_customer after position i
    in routes[route_index].
    """
    best_delta = float('inf')
    best_route, best_pos = None, None
```

Analytical Studies — Dynamic VRP Literature Survey I

Table 11.3 in the textbook (reproduced and summarised below) provides an overview of the major algorithmic approaches for dynamic VRP with requests, drawn from the published literature. The table distinguishes approaches by the type of dynamic VRP addressed and the algorithmic family used.

Summary of key approaches for dynamic requests

Reference	Problem type	Algorithm
Gendreau et al. (1999)	Dynamic VRPTW	Tabu search with double re-insertion
Liao et al. (1992)	Dynamic PDP	Insertion with dispatch rules
Taillard et al. (1996)	Dynamic VRP	Tabu search, adaptive memory
Gendreau & Potvin (1998)	Dynamic VRPTW	Branch & bound with re-optimisation
Bent & Van Hentenryck (2004)	Dynamic VRPTW	Multiple plans, Monte Carlo
Hvattum et al. (2006)	Dynamic VRPTW	Branch & bound per scenario
Wen et al. (2010)	Dynamic VRPTW	Adaptive large neighbourhood search
Goodson et al. (2012)	Dynamic VRPTW	Rollout, DP approximation

Analytical Studies — Dynamic VRP Literature Survey II

The work of Gendreau, Guertin, Potvin, and Seguin (1999) is a landmark in dynamic VRP research. They proposed a **tabu search** algorithm specifically designed for dynamic VRPTW, running continuously in real time. Their double re-insertion move simultaneously relocates two customers, which is more powerful than single-customer re-insertions but still fast enough for online use.

Taillard et al. (1996) introduced the concept of **adaptive memory** programming, where high-quality partial routes found in previous planning cycles are stored and reused in future optimisations. This is particularly effective in rolling horizon settings where routes change only incrementally between epochs. Wen et al. (2010) proposed an **adaptive large neighbourhood search** (ALNS) for dynamic VRPTW. ALNS combines multiple *destroy operators* (which remove a subset of customers from routes) with *repair operators* (which reinsert those customers). The algorithm adaptively selects which operators to use based on their historical performance, making it self-tuning over the course of a single run.

Dynamic Travel Times — Congestion and Variability I

In real-world transportation networks, travel times are not constant. They depend on the **time of day** (due to rush-hour congestion), on **day-of-week effects**, on **weather**, and on **incidents** such as accidents or road closures. Algorithms that ignore this variability and use fixed travel times may produce routes that violate time windows in practice or that are significantly sub-optimal.

Time-Dependent Travel Time Model

Let $t_{ij}(\tau)$ (t sub ij of τ , where τ (τ) is the departure time) be the travel time from node i to node j when departure occurs at time τ . In the simplest model, travel times are piecewise-linear functions of departure time:

$$t_{ij}(\tau) = \frac{d_{ij}}{v_{ij}(\tau)}$$

where d_{ij} is the fixed distance from i to j and $v_{ij}(\tau)$ (v sub ij of τ) is the average speed on that arc at departure time τ .

Dynamic Travel Times — Congestion and Variability II

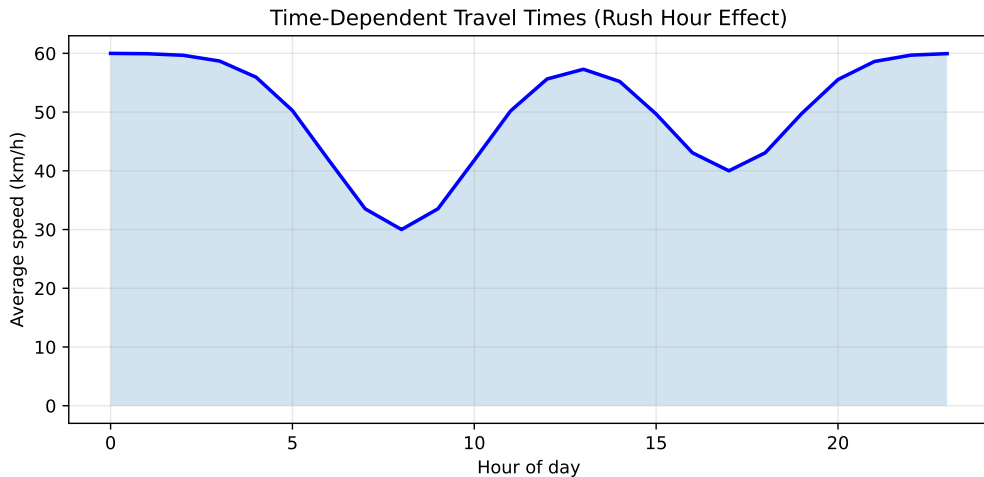
Notation

$t_{ij}(\tau)$ = travel time from i to j when leaving i at time τ ; d_{ij} = fixed geographic distance (km); $v_{ij}(\tau)$ = speed (km/h) on arc (i,j) at departure time τ ; τ (tau, Greek letter) = clock time of departure.

How to read this formula

Think of $t_{ij}(\tau)$ as a travel-time function rather than a single number. If you leave node i at 8 am when rush hour is in full swing, $v_{ij}(8h)$ is low (say, 20 km/h in heavy traffic), so the travel time is high. If you leave at 11 am, $v_{ij}(11h)$ might be 60 km/h, giving one-third the travel time. Routing algorithms that account for this dependency can save substantial time by scheduling stops in a sequence that avoids peak congestion.

Dynamic Travel Times — Congestion and Variability III



Dynamic Travel Times — Congestion and Variability IV

Figure: Illustration of time-dependent travel times for a fixed 30 km arc. Morning and evening rush hours create two congestion peaks, raising the travel time from 30 minutes (free-flow baseline, blue dashed) to nearly 50 minutes. The red shaded area shows the additional delay due to congestion.

Dynamic Travel Times — Congestion and Variability V

FIFO Property (First In, First Out)

A time-dependent travel time function satisfies the **FIFO property** if leaving later never allows you to arrive earlier:

$$\tau_1 < \tau_2 \implies \tau_1 + t_{ij}(\tau_1) \leq \tau_2 + t_{ij}(\tau_2)$$

In words: if vehicle A departs before vehicle B on the same arc, then vehicle A arrives no later than vehicle B. The FIFO property ensures that shortest-path algorithms like Dijkstra's algorithm generalise correctly to time-dependent networks.

Why FIFO matters

Most standard shortest-path algorithms assume that waiting is always free (you can always wait and then travel). If the FIFO property holds, then a vehicle that departs earlier always arrives no later, which means there is never any benefit to deliberately delaying departure. Without the FIFO property, route planning becomes significantly more complex because optimal solutions might involve deliberate waiting at intermediate nodes.

Dynamic Travel Times — Congestion and Variability VI

Several algorithmic approaches for the time-dependent VRP (TDVRP) have been proposed:

Algorithms for TDVRP

- 1 **Adapted Clarke-Wright savings.** The savings algorithm is modified so that the saving s_{ij} for merging two routes at nodes i and j accounts for the time-dependent travel time at the estimated departure time.
- 2 **Tabu search with time-dependent cost evaluation.** Standard VRP tabu search, but whenever a move is evaluated, the route is re-evaluated using $t_{ij}(\tau)$ for the actual departure time at each arc.
- 3 **Evolutionary algorithms.** Genetic algorithms with time-dependent cost functions embedded in the objective function evaluation.

Ichoua, Gendreau, and Potvin (2003) provide a comprehensive study of time-dependent VRP. They model speed as a step function of time (constant speed within each time interval) and show that incorporating time-dependency substantially reduces both total travel time and time-window violations compared with static routing.

Dynamic Travel Times — Congestion and Variability VII

Maden, Eglese, and Black (2010) validated time-dependent VRP approaches using real road network data from the UK, demonstrating practical savings of 5–15% in total travel time when congestion is explicitly modelled.

Huang, Chen, and Chang (2014) proposed a dynamic VRP with both time-dependent travel times and online requests, integrating a congestion model with a rolling-horizon re-optimisation framework. Their computational results confirm that ignoring time dependency leads to systematic route infeasibility during peak hours.

Dynamic Vehicle Availability — Breakdowns and Windows I

A further source of dynamism arises from the vehicles themselves rather than from customers or the road network. In practice, a vehicle may:

- **Break down** mid-route, requiring emergency retrieval and redistribution of its remaining customers to other vehicles.
- **Become available late**, for example because it was used the previous evening and needs to return from a remote depot.
- **Have a restricted time window** during which it may operate, such as when a driver has contractual duty-time limits.

Dynamic Vehicle Availability — Breakdowns and Windows II

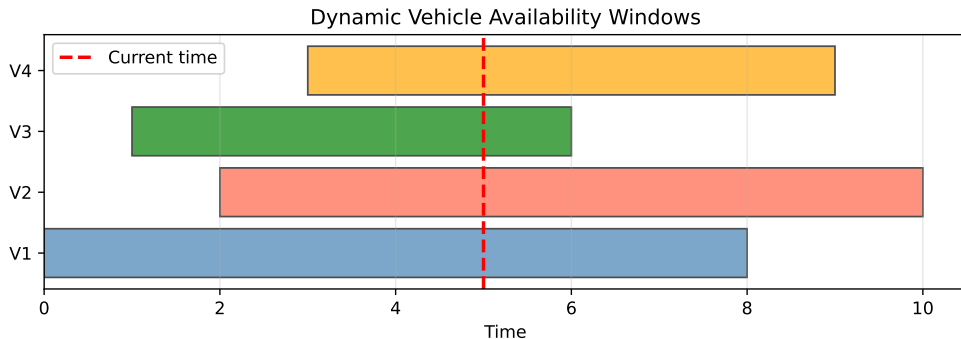


Figure: Dynamic vehicle availability: Vehicle 2 suffers a breakdown from 10:00 to 14:00; Vehicle 4 is not available until 08:00. Algorithms must re-assign the affected customers to other vehicles in real time.

Dynamic Vehicle Availability — Breakdowns and Windows III

Problem Formulation — Vehicle Breakdown

When vehicle k breaks down at time t_b (t sub b) at location ℓ (ℓ , representing a position), the remaining unserved customers U_k in its route must be reassigned. The sub-problem is:

Given: Current positions of all vehicles $k' \neq k$; remaining loads and route plans of vehicles $k' \neq k$; the set of unserved customers U_k from the broken-down vehicle.

Find: An assignment of customers in U_k to vehicles $k' \neq k$ such that:

- Capacity constraints of all vehicles are maintained.
- Time-window constraints of all customers are maintained.
- The total additional distance driven is minimised.

This sub-problem is itself a VRP with fixed starting positions and is solved using the same heuristics as those for dynamic request insertion.

Dynamic Vehicle Availability — Breakdowns and Windows IV

Berbeglia, Cordeau, and Laporte (2010) provide a comprehensive review of vehicle breakdown models in dynamic VRP. They distinguish two sub-cases:

Soft breakdown: The vehicle can continue operating, but at reduced capacity or speed. The problem reduces to re-optimising routes with modified vehicle parameters.

Hard breakdown: The vehicle is completely inoperable. All remaining customers must be reassigned immediately.

Emergency vehicle dispatch as a special case

Emergency vehicle routing (ambulances, fire trucks) combines dynamic requests with dynamic vehicle availability. Calls arrive randomly, must be served within tight time limits, and vehicles may be deployed (temporarily unavailable) for extended periods. Liu, Madden, and Beibel (2004) study this problem and show that preemptive redeployment strategies (repositioning free vehicles anticipatorily) can substantially reduce mean response times.

Key insight

The correct response to a vehicle breakdown in a dynamic VRP depends on how far from the depot the breakdown occurs and how many customers remain unserved. If only one or two customers remain, it may be cheaper to send an entirely new vehicle. If many customers remain and other vehicles are nearby, redistribution is preferred. A fast feasibility check and a cheapest-insertion heuristic are typically sufficient for real-time response.

Vehicle Re-routing after Breakdown — Algorithm I

Re-routing Algorithm after Breakdown

Input: Broken-down vehicle k ; remaining customer set U_k ; current positions and plans of active vehicles $k' \in K \setminus \{k\}$.

Output: Updated route plans for all active vehicles.

- ① Sort customers in U_k by urgency: those with tight time windows first.
- ② **For each** customer $c \in U_k$ in priority order:
 - ① Apply cheapest feasible insertion across all active vehicles.
 - ② If insertion is feasible: add c to that vehicle's plan.
 - ③ If no feasible insertion exists:
 - Attempt rerouting (remove a less-urgent customer from another route to make room).
 - If still infeasible: flag c as unserved and alert dispatcher.
- ③ Re-optimize each affected route using a local search (e.g., 2-opt) to recover any unnecessary detours introduced by insertions.

Vehicle Re-routing after Breakdown — Algorithm II

Practical note

Step 3 (local search improvement) is optional but important. Simple cheapest insertion can produce routes with unnecessary “zigzags” because each insertion is made independently. Running a quick 2-opt improvement pass after all insertions typically reduces total distance by 5–15%.

Benchmarking Dynamic VRP Algorithms I

Measuring the quality of a dynamic VRP algorithm is considerably more complex than measuring a static VRP algorithm. In a static VRP, a single optimal solution (or a known lower bound) can serve as the benchmark. In a dynamic VRP, the set of customers is not known at the start of execution, making a direct comparison to an “optimal” solution difficult.

The Offline Optimal as a Benchmark

The most common benchmark for dynamic VRP algorithms is the **offline optimal**: the cost of the optimal solution to the *static* VRP instance that includes *all* customers — both a priori and dynamic — as if they had all been known in advance.

Formally, let Z_{offline}^* be the optimal cost when all customer locations are known from the start. Let $Z_{\text{online}}(\mathcal{A})$ be the cost achieved by online algorithm $\mathcal{A}$ that sees requests as they arrive. The **online ratio** (or *competitive ratio*) of algorithm $\mathcal{A}$ is:

$$\rho(\mathcal{A}) = \frac{Z_{\text{online}}(\mathcal{A})}{Z_{\text{offline}}^*}$$

Benchmarking Dynamic VRP Algorithms II

Notation

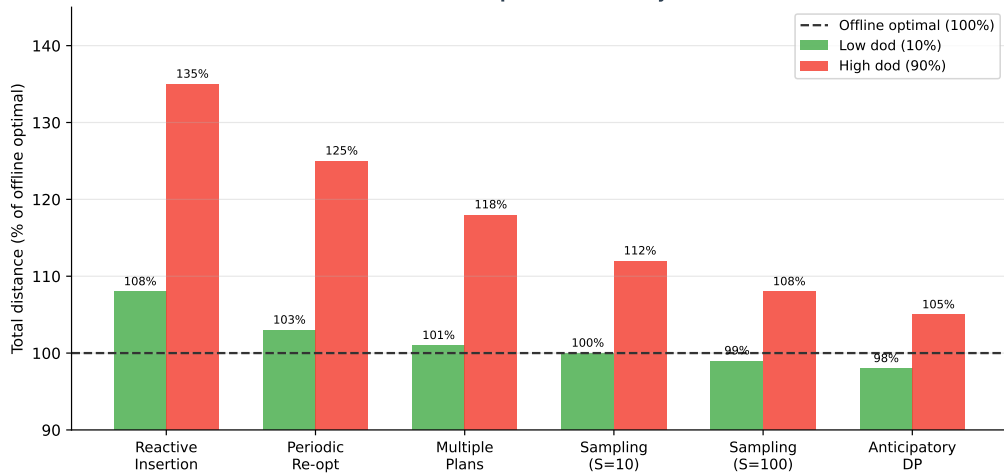
Z_{offline}^* = minimum cost achievable with perfect foresight; $Z_{\text{online}}(\mathcal{A})$ = actual cost achieved by algorithm $\mathcal{A}$ under online conditions; $\rho(\mathcal{A})$ (rho, the Greek letter) = ratio of online to offline cost. A value of $\rho = 1$ means the online algorithm performs as well as perfect-foresight planning (impossible in practice); a value of $\rho = 1.2$ means the online algorithm uses 20% more distance than the theoretical offline optimum.

How to interpret ρ

The online ratio ρ is always ≥ 1 because the algorithm faces more uncertainty than the offline optimum. Values of ρ near 1 are desirable. For highly dynamic problems (high dod), ρ values of 1.2–1.4 are typical even for the best known algorithms. For problems with low dod, good anticipatory algorithms can achieve ρ very close to 1.

Benchmarking Dynamic VRP Algorithms III

Illustrative Performance Comparison across Dynamic VRP Methods



Benchmarking Dynamic VRP Algorithms IV

Figure: Illustrative performance comparison of dynamic VRP solution methods across two settings: low degree of dynamism (10% dynamic requests, green bars) and high degree of dynamism (90% dynamic requests, red bars). All values are expressed as percentages of the offline optimal (100%). Sophisticated anticipatory methods offer larger advantages at low dod; at high dod the gap narrows.

Benchmarking Dynamic VRP Algorithms V

Simulation-Based Evaluation

Because dynamic VRP performance depends on the realisation of random arrivals, a single-run comparison is unreliable. Best practice is to evaluate algorithms over many *randomly generated scenarios* using a consistent simulation framework.

A simulation study typically proceeds as follows:

- 1 Fix a spatial distribution for customer locations (e.g., uniform on a $[0, 100]^2$ square, or clustered around m centres).
- 2 Fix an arrival-time distribution (e.g., Poisson with rate λ (lambda, read “lambda”) requests per hour).
- 3 Generate N_{sim} independent scenario realisations.
- 4 For each realisation, run the dynamic VRP algorithm and record the cost $Z_{\text{online}}^{(r)}$ for replicate r .
- 5 Report the mean $\bar{Z} = \frac{1}{N_{\text{sim}}} \sum_r Z_{\text{online}}^{(r)}$ and a 95% confidence interval.

Larsen, Madsen, and Solomon (2002, 2004) standardised the evaluation framework for dynamic VRPTW, proposing benchmark instance families based on Solomon’s static VRPTW instances with additional

dynamic components. These benchmarks have been widely adopted in subsequent research, making results comparable across different papers and research groups.

Tables from the Textbook — Algorithmic Overview I

Tables 11.1 through 11.4 in the textbook systematically classify over 60 papers on dynamic VRP. The following summarises key dimensions of the classification:

Dimensions of the classification

Dimension	Values / categories
Problem type	DVRP, D-VRPTW, D-DARP, D-PDP, emergency routing
Objective	Min distance, min vehicles, min rejections, min lateness
load range	Low (<30%), medium (30–70%), high (>70%)
Algorithm	Insertion, tabu search, GA, DP, sampling, ALNS
Waiting policy	No-wait, wait, optimal-wait
Benchmark	Larsen instances, random instances, real-world data

Key observations from the survey:

- 1 The majority of papers focus on the dynamic VRPTW (with time windows), reflecting the practical importance of customer time-window constraints in delivery operations.

Tables from the Textbook — Algorithmic Overview II

- ② Tabu search and its variants dominate the metaheuristic approaches, largely because tabu search can be interrupted and restarted cheaply between events.
- ③ Sampling-based anticipatory methods consistently outperform purely reactive methods when the degree of dynamism is low to medium.
- ④ At high dod values, reactive insertion with a good local search post-processor is often competitive with sophisticated anticipatory methods, because there is little advance information to exploit.

Python — Simulation Framework for Dynamic VRP Evaluation

The following code implements a minimal simulation framework for evaluating a dynamic VRP algorithm. Requests arrive according to a Poisson process. The algorithm is called each time a new request arrives and whenever a vehicle becomes idle.

```
import random, math, heapq
from typing import Callable, List, Tuple

#         Simulation event types

REQUEST_ARRIVAL = 'request'
VEHICLE_IDLE     = 'idle'

def poisson_arrivals(rate: float, horizon: float, seed: int = 42):
    """
    Generate Poisson-distributed request arrival times over [0, horizon].
    rate = expected number of requests per unit time.
    Returns sorted list of arrival times.
    """
    rng = random.Random(seed)
    times, t = [], 0.0
    while True:
        inter = -math.log(rng.random()) / rate    # exponential inter-arrival
        t += inter
        if t > horizon:
            break
        times.append(t)
```

Conclusions — What We Have Learned I

Chapter 11 has surveyed the field of dynamic vehicle routing problems, covering the three main sources of dynamism (requests, travel times, vehicle availability) and the corresponding algorithmic approaches.

Conclusions — What We Have Learned II

Summary of main findings

- ① **Dynamic requests** are the most studied form of dynamism. Reactive insertion heuristics are fast and practical; periodic re-optimisation with tabu search or ALNS provides better solution quality; sampling-based anticipatory methods outperform reactive approaches when a reliable probabilistic model of future arrivals is available.
- ② **Dynamic travel times** can be addressed by incorporating time-dependent cost functions into standard routing heuristics. The FIFO property is a key theoretical requirement for these models. Practical savings of 5–15% in total travel time are reported in real-world studies.
- ③ **Dynamic vehicle availability** (breakdowns, windows) is handled by re-running insertion and local search procedures upon each disruption event. Preemptive redeployment (anticipating future availability changes) can reduce response times in emergency settings.
- ④ **Evaluation** of dynamic VRP algorithms requires simulation over many random scenarios. The offline optimal provides a universal benchmark. The degree of dynamism (dod) is the key problem characteristic that determines which algorithmic family performs best.

Conclusions — What We Have Learned III

Central takeaway

There is no single best algorithm for dynamic VRP. The optimal choice depends on the degree of dynamism, the available computation time between events, and the accuracy of the probabilistic model of future arrivals. For high-dod, time-critical settings: use fast reactive insertion. For low-to-medium dod with adequate compute: use anticipatory sampling. For problems where routes change incrementally: use adaptive memory and warm-started metaheuristics to exploit solution continuity.

Future Research Directions I

The chapter closes by identifying several directions in which the field is likely to develop:

Future Research Directions II

Open research directions

- ④ **Development of tighter solution frameworks.** Most current dynamic VRP algorithms are purely heuristic. Developing lower bounds and exact sub-problem solvers that can be embedded in online algorithms would improve the ability to assess solution quality and guide search.
- ② **Improving and extending anticipation schemes.** Current anticipatory algorithms use relatively simple probabilistic models of future arrivals. Richer models — including spatial correlations, demand clustering, and time-of-day patterns — could improve anticipation significantly, particularly for urban logistics applications.
- ③ **Dynamic real-time performance.** As vehicle-to-infrastructure communication improves, real-time traffic data become available at sub-minute resolution. Algorithms that update travel time estimates continuously and re-route vehicles accordingly (rather than at fixed planning epochs) represent an important frontier.
- ④ **Emergency services.** The field of dynamic routing for emergency services (ambulances, police, fire) remains underdeveloped relative to its practical importance. The combination of dynamic requests, strict time constraints, and dynamic vehicle availability creates unique algorithmic challenges.
- ⑤ **Parallel algorithms and big-data settings.** Modern parcel-delivery and ride-hailing platforms may process thousands of requests per minute across fleets of hundreds of vehicles. Parallel and

Selected Bibliography I

The following references are central to the material covered in this chapter. They are listed in order of their significance to the topics discussed.



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Introduced the degree of dynamism (dod) measure and the first systematic benchmark study.

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Chapter 11 — Dynamic Vehicle Routing Problems

Vehicle Routing: Problems, Methods, and Applications
2nd Edition, 2014

Chapter in one sentence

Dynamic VRP extends classical routing to handle real-time changes in requests, travel times, and vehicle availability; effective algorithms combine fast reactive heuristics with anticipatory methods that model the uncertainty of future events.